

One Prefill Prices a Pool of Reasoning Models

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Abstract

Reasoning-model routers must estimate generation cost before choosing a model, yet output lengths vary substantially across queries. We predict costs across a reasoning-model pool from one frozen 4B encoder’s prefill activations. Per-route linear cost readouts reuse the representation used for success prediction, without requiring target-model activations or generation before dispatch. Holding success predictions fixed, query-dependent pricing saves 35.6% on LiveCodeBench, 21.9% on Omni-MATH-500, and 30.7% on MMLU-Pro relative to median-length pricing at matched accuracy. Adapted embedding and prompt-feature estimators capture less of this value, with inconclusive differences on Omni. On MMLU-Pro, dedicated cost readouts add 21.5 percentage points over pricing derived from predicted success. Other coding pools have oracle headroom but no significant learned gain. These results support shared-prefill cost prediction as a routing component whose value depends on the workload.

1. Introduction

Reasoning-model routing requires predicting both correctness and generation cost before choosing a model. Token rates are known, but reasoning-trace lengths depend on the query. A model that is cheap on average may be expensive on a particular problem. Query-dependent pricing can therefore change which route offers the best trade-off.

Prefill routers estimate correctness from prompt activations. The prefill-router architecture of Varshney et al. [8] explicitly defers output-length prediction and prices outputs using median training length. We investigate whether the same kind of representation can also predict output lengths across a pool of reasoning models. One frozen 4B encoder processes the query; per-route linear readouts estimate success and cost before dispatch (Fig. 1). Cost prediction adds readouts to an existing prefill router without another encoder pass.

We evaluate this cost signal by holding success predictions and the routing objective fixed while changing pricing. Three workloads yield 22–36% savings against median-length pricing. A dedicated readout is especially useful on

MMLU-Pro, where predicted success leaves useful cost information unexplained. Oracle headroom and negative coding results establish the method’s limits.

Related work. Learned cost is established in routing: MixLLM predicts output length from query embeddings [9], and CARROT estimates cost and performance using embedding-based nearest neighbors or a shared fine-tuned RoBERTa encoder [7]. Activation-based length prediction also supports inference scheduling [6, 11]. Our focus is frozen generative prefill features for pricing multiple reasoning routes through simple readouts. ZeroRouter derives length from item-response difficulty bins [12]; we examine whether pricing benefits from information beyond predicted success.

2. Cross-Model Cost from One Prefill

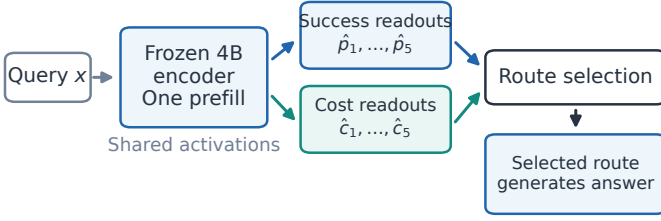
For query x and route m , let $p_m(x)$ be the probability of correctness and $c_m(x)$ the expected generation cost. A route is a model with a fixed reasoning-effort setting. We select one route before generation:

$$m^*(x; V) = \arg \max_m \{V \hat{p}_m(x) - \hat{c}_m(x)\}, \quad (1)$$

where V sets the value of a correct answer. There is no verifier or fallback call.

Shared features, separate readouts. We concatenate mean-pooled and last-token activations from eight stored layers of a frozen Qwen3-4B encoder and standardize using training statistics. The Instruct-2507 encoder is used on LiveCodeBench and MMLU-Pro; Thinking-2507 is fixed for Omni. Both readouts are regularized linear models. Logistic success heads use a binomial likelihood over valid draws and held-out Platt calibration. Ridge cost heads fit squared error in log mean output length, with training-only cross-validation for regularization. The different losses match correctness outcomes and continuous lengths. Success-head penalty selection and Platt calibration use the first draw. Each cost head needs output-length labels from its target route; zero-shot transfer to unseen models is not assumed.

(a) One encoding prices the entire pool



Target activations and generations are unnecessary before dispatch.

Against median-length pricing at matched accuracy; whiskers: paired 95% bootstrap CIs.

(b) Same success head, better pricing

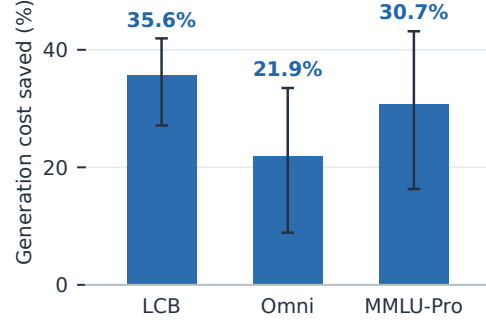


Figure 1. **One prefill. Five prices. One generation.** (a) A frozen encoder supplies shared features to per-route success and cost readouts before one route is selected. Each route supplies training labels, but its activations and generation are unnecessary before dispatch. (b) Holding success predictions fixed, predicted costs reduce generation spend relative to median-length pricing at matched accuracy. Whiskers show paired 95% bootstrap intervals; common encoder overhead is excluded.

From length to price. We exponentiate predicted log length, apply training-residual smearing, and match the route’s training mean. For predicted output tokens $\hat{\ell}_m(x)$, input tokens $n_m^{\text{in}}(x)$, and input/output rates in dollars per million,

$$\hat{c}_m(x) = \frac{r_m^{\text{in}} n_m^{\text{in}}(x) + r_m^{\text{out}} \hat{\ell}_m(x)}{10^6}. \quad (2)$$

The reference substitutes median valid training output length. The empirical oracle substitutes mean observed problem–route output length. Only the oracle uses evaluation lengths when selecting a route; every arm is charged observed tokens after its decisions are fixed.

3. Experimental Protocol

The main pools contain 892 LiveCodeBench problems [3], all 500 Omni-MATH-500 problems from Omni-MATH [1], and a subject-stratified 1,000-question MMLU-Pro subset [10]. Train/calibration/test counts are 441/110/341, 275/75/150, and 550/150/300. The five routes are gpt-oss-20b at low/medium effort, deepseek-v4-flash, and gpt-oss-120b at medium/high effort. Approximately 16/12/8/6/3 valid draws per route are retained on LiveCodeBench and 4/3/3/2/2 on the other pools. Correctness and cost are estimated from all valid draws; generations are not treated as independent test problems.

Pricing and controls. We hold the recorded market rates fixed across arms: input/output dollars per million are 0.018/0.09 for gpt-oss-20b, 0.04704/0.09408 for deepseek-v4-flash, and 0.15/0.60 for gpt-oss-120b. Reported savings

isolate target generation spend. The encoder pass is common to success-only and success-plus-cost routing, so these are incremental savings for an existing prefill router. Training and calibration exclude test problems.

Matched-accuracy evaluation. We sweep V , construct test accuracy–cost frontiers, and allow randomized mixtures between convex-hull points. Let $C_{\text{arm}}(a)$ denote cost at accuracy a . Savings against median pricing are

$$G = 1 - \exp \left[\frac{1}{12} \sum_{j=1}^{12} \log \frac{C_{\text{arm}}(a_j)}{C_{\text{median}}(a_j)} \right], \quad (3)$$

using 12 targets over the interior 5–95% of the accuracy band shared by the estimator, median reference, and empirical oracle. Each estimator’s band can differ; paired differences below compare these savings summaries. Oracle G measures available headroom. We use 500 paired problem bootstrap resamples, keeping all routes and draws of a problem together. Intervals condition on fitted heads. These descriptive frontiers do not measure a deployment operating point chosen before evaluation. The separate end-to-end ZeroRouter comparison uses 1,000 resamples and its own comparison bands.

4. Results

One prefill provides useful cross-model prices. Changing only pricing saves 35.6% on LiveCodeBench, 21.9% on Omni, and 30.7% on MMLU-Pro; all three intervals exclude zero (Table 1). These gains capture approximately 78%, 49%, and 47% of empirical oracle headroom (46.0%, 44.2%, and 65.1%).

Cost estimator	LiveCodeBench	Omni-MATH-500	MMLU-Pro
Embedding ensemble (MixLLM-style)	14.1 [5.3, 21.0]	12.5 [−2.4, 25.6]	4.0 [−4.7, 14.9]
Prompt-feature GBM	12.3 [0.7, 21.4]	6.6 [−12.2, 21.2]	9.8 [−4.5, 25.9]
Cost from success predictions	34.1 [24.7, 40.9]	21.0 [6.6, 33.7]	9.2 [−2.2, 24.6]
Shared-prefill cost readouts (ours)	35.6 [27.1, 41.9]	21.9 [8.9, 33.5]	30.7 [16.3, 43.2]

Table 1. **Change pricing, hold success fixed.** Cost savings G (%) against median-length pricing; brackets give 95% paired-problem bootstrap intervals for each estimator. Bands are defined per estimator as described in the protocol; these are not percentage savings directly against another learned estimator. Embedding and prompt-feature baselines are adaptations. Significance of estimator differences requires the paired contrasts reported in the text.

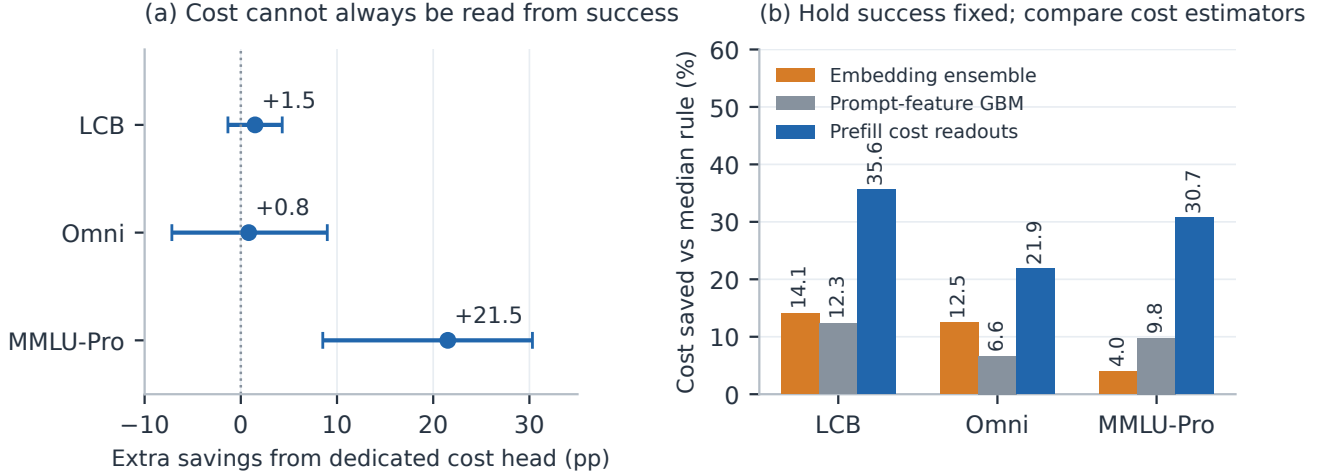


Figure 2. **Cost information beyond success.** (a) Added savings from dedicated cost readouts over cost inferred from success predictions, with paired 95% intervals. (b) Estimator savings with success predictions held fixed. Panels report percentage-point differences and absolute percentage savings, respectively. Bars in (b) are point estimates; Table 1 gives intervals.

The embedding baseline averages per-route MLP, random-forest, and nearest-neighbor regressors on jina-code embeddings, adapting MixLLM’s predictor family. The GBM uses prompt length, numbers, examples, constraints, and keyword counts. Both fit log output length using the same training split and price conversion. On MMLU-Pro, paired advantages over these estimators are 26.7 points [12.4, 40.2] and 20.9 [3.8, 33.3]. On Omni, the differences are inconclusive: 9.4 [−4.4, 24.2] and 15.2 [−1.4, 34.1]. These adaptations do not isolate representation choice from encoder size and regression design.

Does cost need information beyond success? We regress log output length on the success heads’ predicted logits and their squares, keeping success predictions fixed in routing. This control saves 34.1% on LiveCodeBench and 21.0% on Omni, with inconclusive differences from dedicated cost readouts (Fig. 2). On MMLU-Pro it saves only 9.2%, versus 30.7% for the dedicated head: a paired advantage of 21.5 points [8.5, 30.3].

On MMLU-Pro, empirical outcome-based difficulty explains little of log length (mean $R^2 \approx .19$). Adding subject

to success-derived pricing raises savings to 22.3%, recovering roughly 60% of its gap to the full cost probe. Domain variation supplies useful cost information; required work is a plausible interpretation rather than a measured causal variable.

Compatibility with another success architecture. Our ZeroRouter reimplementation uses its item-response model and difficulty-bin pricing [12], with MAP instead of SVI and frozen 4B features for latent prediction. Dimension, seed, and bin count are selected on calibration. Our end-to-end advantage is 17.5 points [9.8, 24.9] on LiveCodeBench, with inconclusive differences on MMLU-Pro (+9.3 [−2.6, 21.5]) and Omni (+2.8 [−10.7, 16.6]). Holding its success predictions fixed, substituting our cost heads raises savings from 18.5 to 35.5%, 20.0 to 22.0%, and 21.6 to 32.4%, respectively. These descriptive swaps show that our pricing composes with a second success architecture; this is a deployment-pool reimplementation rather than its original population-scale experiment.

Headroom does not ensure a learned improvement. CodeContests [5], TACO [4], and BigCodeBench [13] have oracle headroom of 21.8%, 36.1%, and 20.2%, yet learned gains of 2.0%, 2.1%, and -3.3% have intervals spanning zero. RouterBench’s chat pool [2] has only 10.5% headroom. These contrast pools differ in models, prices, and accuracy ladders. They demonstrate workload-dependent limits, not a controlled causal effect of reasoning. Additional controls and the headroom plot appear in the supplement.

5. Discussion and Limitations

One encoder can supply prices for several target models, but per-route cost labels remain necessary. Our pools cover five configurations from two model families. The adapted estimators do not reproduce complete MixLLM or CARROT routers, and CARROT’s exact estimators are not evaluated. Bootstrap intervals omit training and calibration-selection uncertainty; empirical oracle lengths also have sampling error. Replay evaluates stored generations, and matched-accuracy frontiers allow mixtures chosen using evaluation outcomes. Encoder overhead is excluded from incremental generation savings. Larger held-out samples, broader model families, and calibration-selected operating points would strengthen the evidence.

6. Conclusion

One frozen prefill can supply success and cost predictions for a reasoning-model pool. Linear cost readouts save 22–36% on three workloads while reusing the success encoder. Dedicated cost information is particularly useful on heterogeneous MMLU-Pro, where success-derived pricing captures less of the available value. Substantial oracle headroom can remain uncaptured on other workloads. Shared-prefill cost prediction offers a practical addition to reasoning-model routing with explicit empirical limits.

References

- [1] Bofei Gao, Feifan Song, Zhe Yang, et al. Omni-MATH: A Universal Olympiad Level Mathematic Benchmark For Large Language Models, 2024. arXiv:2410.07985.
- [2] Qitian Jason Hu, Jacob Bieker, Xiuyu Li, et al. RouterBench: A Benchmark for Multi-LLM Routing System, 2024. arXiv:2403.12031.
- [3] Naman Jain, King Han, Alex Gu, et al. LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code, 2024. arXiv:2403.07974.
- [4] Rongao Li, Jie Fu, Bo-Wen Zhang, et al. TACO: Topics in Algorithmic COde generation dataset, 2023. arXiv:2312.14852.
- [5] Yujia Li, David Choi, Junyoung Chung, et al. Competition-Level Code Generation with AlphaCode, 2022. arXiv:2203.07814.
- [6] Glen Messenger. ALPS: Activation-Based Length Prediction for Intelligent LLM Inference Scheduling, 2026. Zenodo. doi:10.5281/zenodo.19078431.
- [7] Seamus Somerstep, Felipe Maia Polo, Allysson Flavio Melo de Oliveira, et al. CARROT: A Cost Aware Rate Optimal Router, 2025. arXiv:2502.03261.
- [8] Tanay Varshney, Annie Surla, Michelle Xu, et al. LLM Router: Rethinking Routing with Prefill Activations, 2026. arXiv:2603.20895.
- [9] Xinyuan Wang, Yanchi Liu, Wei Cheng, et al. MixLLM: Dynamic Routing in Mixed Large Language Models, 2025. arXiv:2502.18482.
- [10] Yubo Wang, Xueguang Ma, Ge Zhang, et al. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark, 2024. arXiv:2406.01574.
- [11] Huanyi Xie, Yubin Chen, Liangyu Wang, et al. Predicting LLM Output Length via Entropy-Guided Representations, 2026. arXiv:2602.11812.
- [12] Cheng Yan, Wuyang Zhang, Zhiyuan Ning, et al. Breaking Model Lock-in: Cost-Efficient Zero-Shot LLM Routing via a Universal Latent Space, 2026. arXiv:2601.06220.
- [13] Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, et al. BigCodeBench: Benchmarking Code Generation with Diverse Function Calls and Complex Instructions, 2025. arXiv:2406.15877.